



CIT121 – COMPUTER ARCHITECTURE PROJECT

AI-Powered Healthcare Diagnostic System & Future Architecture

Your Name • Group Members • University Name



Overview of System

An AI-powered healthcare diagnostic system designed to help hospitals process complex medical data faster and with greater accuracy — transforming how diagnosis is performed.



MRI Scans

High-resolution brain and tissue imaging processed in real time



CT Scans

Cross-sectional imaging analyzed for anomalies and pathology



ECG / Real-Time Data

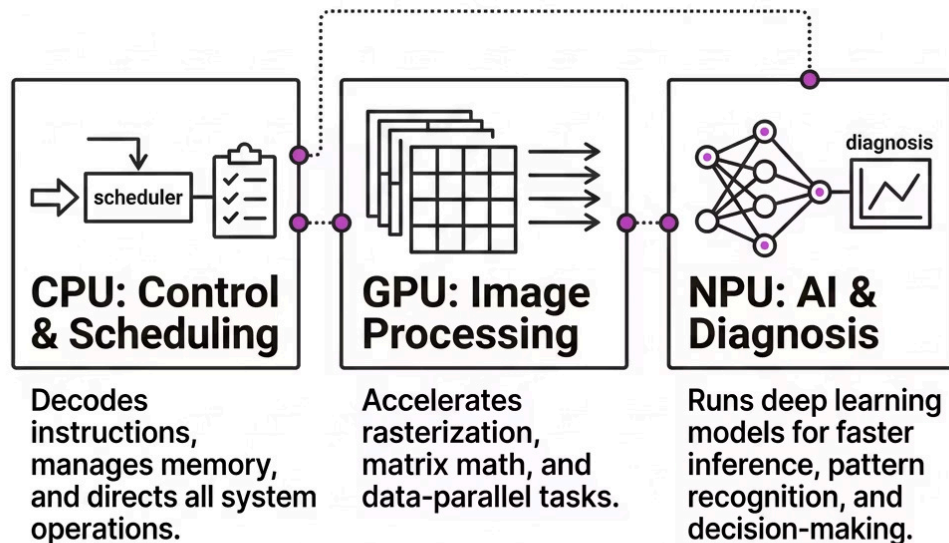
Continuous patient vitals monitored and interpreted by AI



Goal: Faster and more accurate diagnosis using AI-driven processing pipelines.

Heterogeneous Computing System

The system uses **multiple specialized processors**, each optimized for a distinct role — creating a powerful, efficient pipeline for medical AI.



CPU — Control & Scheduling

Orchestrates task flow and manages system resources

GPU — Image Processing

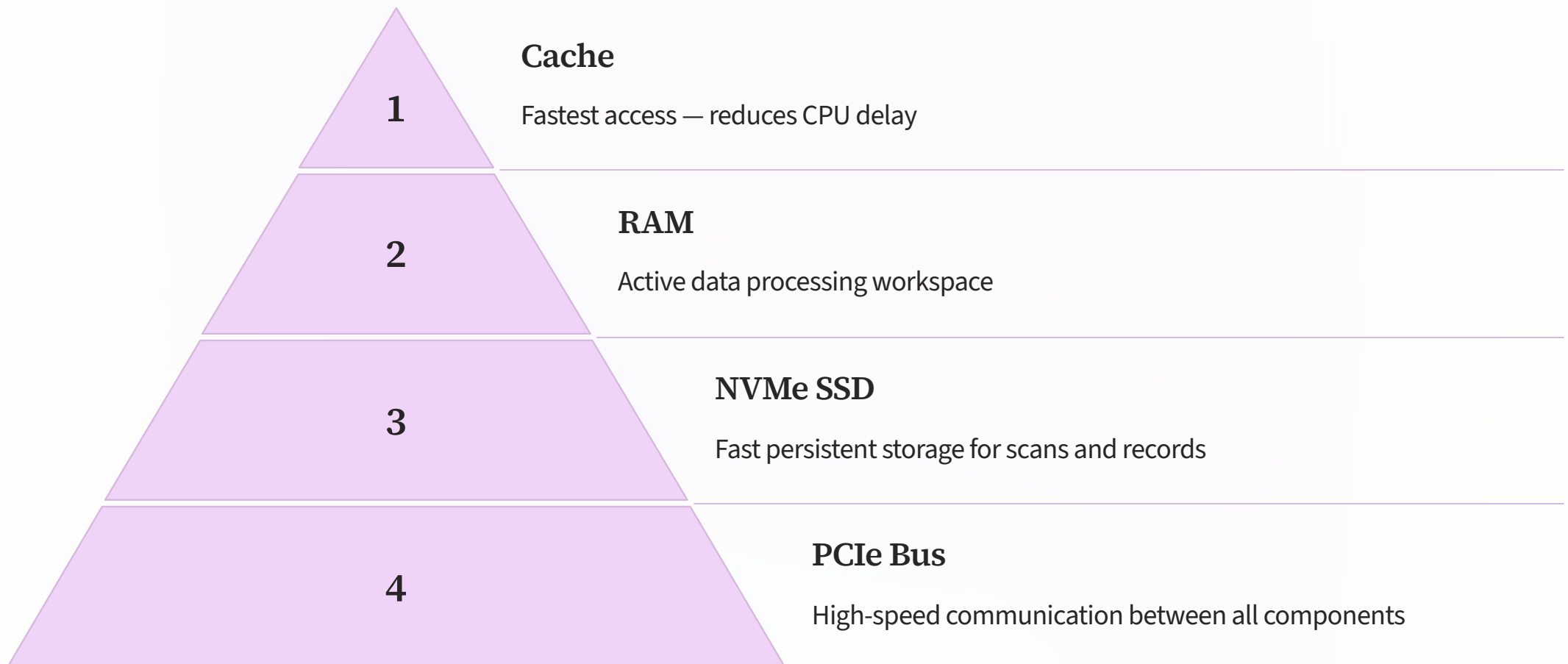
Best for matrix operations — massive parallel processing advantage for medical imaging

TPU/NPU — AI Prediction

Dedicated AI accelerator for neural network inference and diagnosis

Memory Hierarchy

A layered memory architecture ensures the right data is available at the right speed — minimizing latency at every stage of the diagnostic pipeline.



The **PCIe interconnect** serves as the backbone, enabling high-bandwidth data transfer between CPU, GPU, TPU, and storage.

How the System Works

A seamless, automated pipeline moves medical data from acquisition to AI-powered diagnosis — with minimal human intervention and maximum speed.

01

Data Acquisition

Medical data collected from MRI and ECG devices

02

RAM Storage

Raw data stored in RAM for active processing

03

CPU Scheduling

CPU schedules and coordinates all processing tasks

04

GPU Processing

DMA transfers data to GPU for image processing

05

AI Diagnosis

TPU performs AI diagnosis such as tumor detection

06

Results Delivery

Diagnosis results stored and presented to doctors

Processing Division

Each processor type handles the workload it's best suited for — eliminating bottlenecks and maximizing throughput across the entire system.

⚙️ CPU — Control Unit

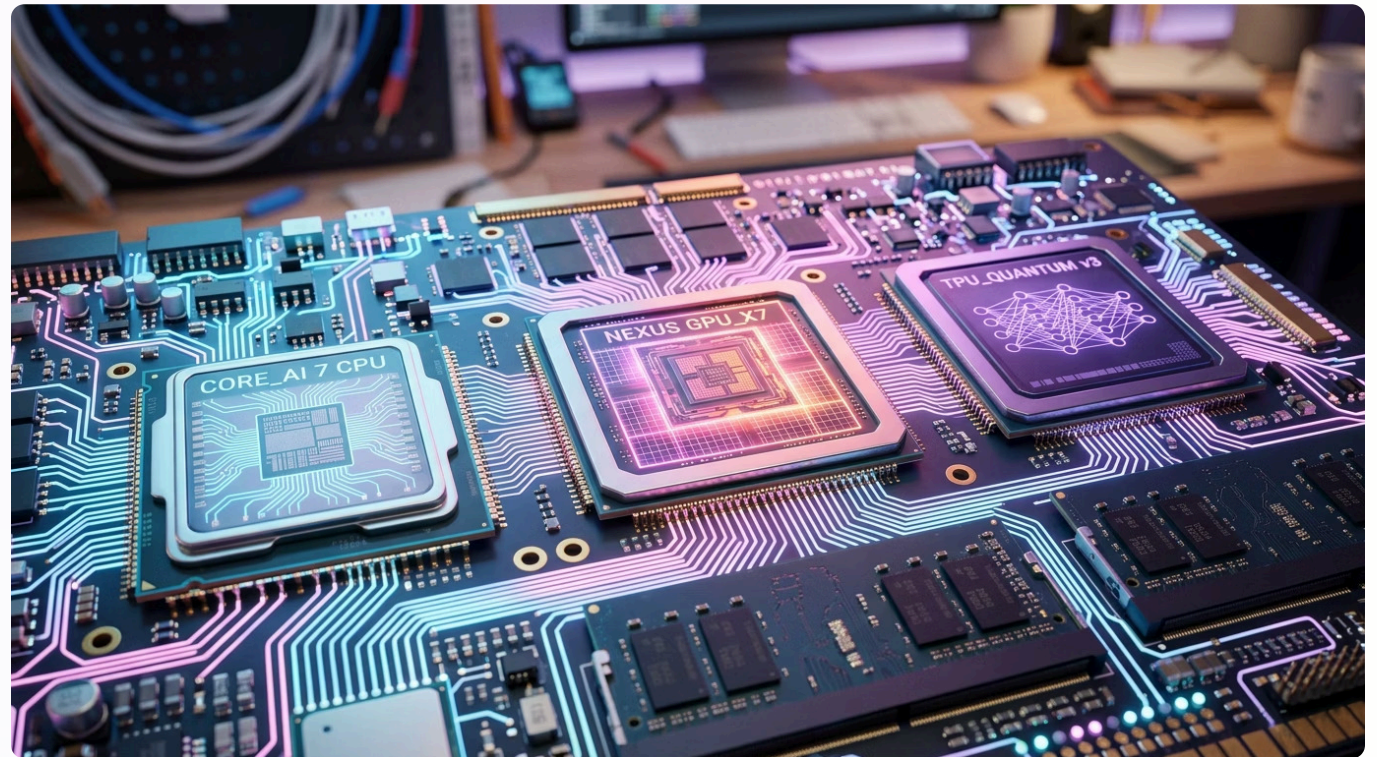
Manages instruction flow, task scheduling, and system coordination

🖥️ GPU — Image Processing

Handles heavy parallel workloads — MRI, CT, and X-ray analysis

🧠 TPU/NPU — AI Prediction

Runs neural network models for tumor detection and anomaly classification



3x

Processor Types

CPU, GPU, and TPU working in harmony

15–30x

Faster Diagnosis

Compared to traditional CPU-only systems

Next-Generation System — 2035 Architecture

By 2035, AI accelerators will become the **primary processors**, with neuromorphic and edge-cloud hybrid architectures redefining medical computing.

1

AI Accelerators

TPUs become the main processing unit for real-time diagnosis

2

Neuromorphic Chips

Brain-inspired, event-driven processors added to the architecture

3

Edge Computing

Local processing at the point of care for instant response

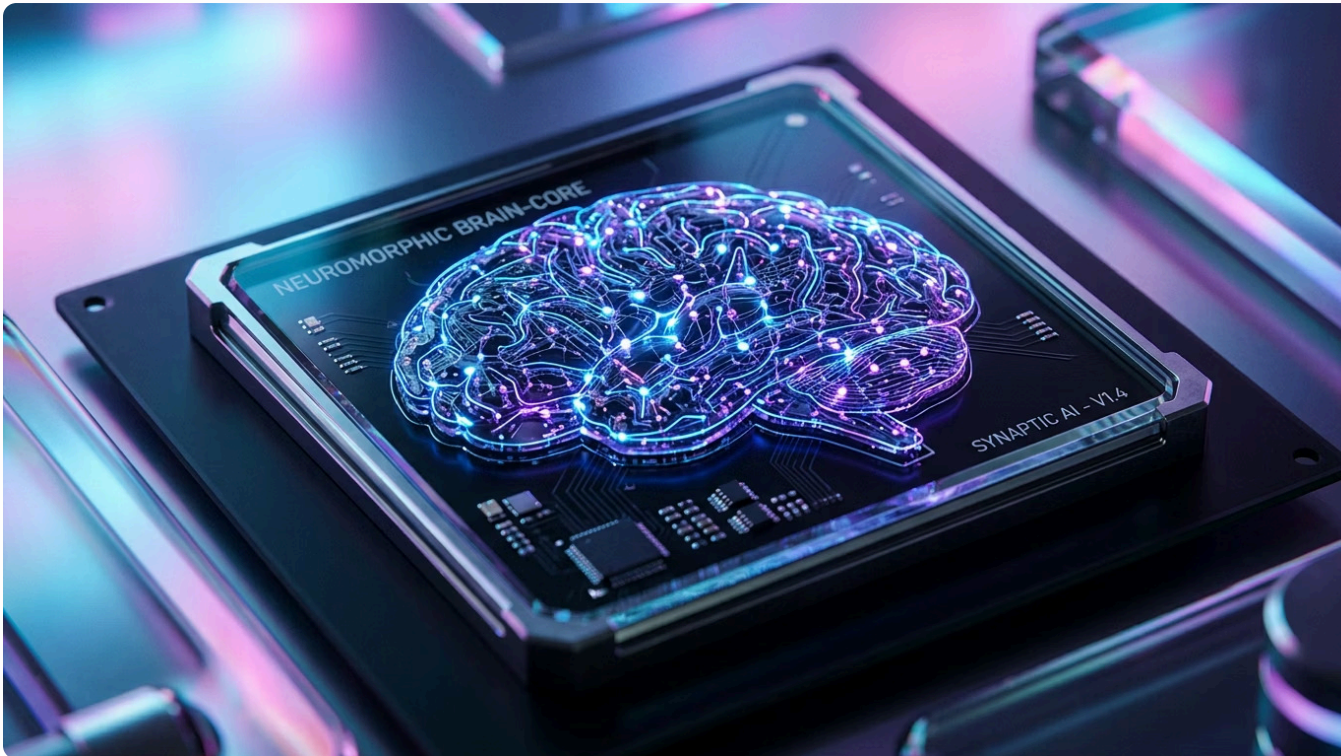
4

Cloud Integration

Centralized storage and large-scale AI model training



Advanced Processing Models



Neuromorphic Processors

Inspired by the human brain, these chips **activate only when data changes** — dramatically reducing power consumption while maintaining high performance.

- Event-driven processing architecture
- Ultra-low power consumption
- Ideal for ECG anomaly detection

Edge Computing

Processing data **locally at the device** eliminates cloud latency — critical for emergency diagnostics where every millisecond counts.

- Faster response in emergencies
- Reduces cloud dependency and delay
- Works even with limited connectivity



System Benefits

The heterogeneous architecture directly addresses the core limitations of traditional computing in medical environments.



Speed

Faster diagnosis with **15–30x improvement** over CPU-only systems



Accuracy

Better AI predictions through specialized TPU/NPU inference engines



Scalability

Easy system expansion as hospital data volumes grow



Solves Von Neumann Bottleneck

Separate memory and processing paths eliminate the classic speed gap



Reduces High Latency

Parallel GPU/TPU processing dramatically cuts wait times



Improves Scalability

Heterogeneous design scales with growing hospital demands

Key Concepts & Conclusion

Cache Memory

Bridges the CPU–RAM speed gap. A **cache hit** means fast access; a **cache miss** falls back to slower RAM.

- **Temporal Locality** — Recently used data is reused
- **Spatial Locality** — Nearby data is accessed together

RISC vs CISC

- **RISC** — Simple instructions, fast execution, power efficient
- **CISC** — Complex instructions, fewer lines of code

Conclusion

Combining AI with advanced computer architecture creates healthcare systems that are **faster, smarter, and more scalable** than ever before.

- AI accelerators transform diagnostic speed
- Neuromorphic and edge computing define the future
- Heterogeneous design solves legacy bottlenecks



Future systems will be faster, smarter, and fully scalable — revolutionizing patient care.